Descriptor Similarity Does Not Predict Array Response: Chemical Warfare Agent Simulants Tested Against Seventeen Authentic Agents

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Abstract

Chemical warfare agent detection is developed and qualified on simulants, and we asked what measurement establishes that a simulant reproduces the agent it stands in for. A 29-dye colorimetric array was used to measure seventeen authentic agents, spanning nerve, blood, choking, vesicant and A-series compounds, at five concentrations in two optical channels, and the resulting response patterns were compared with the molecular-descriptor distances on which simulant selection in practice rests. Agreement was tested by exact Mantel permutation within each run, by exact enumeration of all 680 possible simulant designations in descriptor space, and against permutation nulls built for each comparison. The array itself resolves these agents, at a cross-validated classification accuracy of 0.964 on the seven-agent run and 1.000 on the Novichok run,

so the measurement is sound. Descriptor distance nonetheless fails to predict response distance: twelve exact Mantel tests and six combinations of them leave nothing below $q = 0.05$. Dimethyl methylphosphonate, the standard G-agent simulant and the benchmark compound for decontamination efficiency, is anti-correlated with soman and cyclosarin in the ultraviolet channel, and its correlation with sarin falls from $+0.77$ at 500 μM to $+0.36$ at 10 μM, the end at which point detectors operate. Across the seventeen agents measured, exactly one designated simulant has ever been measured beside them.

Introduction

Nearly every chemical warfare agent detector in the literature is developed, optimised and qualified on compounds that are not chemical warfare agents.[1,2] Dimethyl methylphosphonate stands in for the G-series nerve agents,[3] dipropylene glycol monomethyl ether for training detectors,[4] and methyl salicylate for vapour penetration through protective ensembles.[5,6] The substitution is unavoidable, because the agents are lethal at concentrations where handling them requires a licensed facility, and it is load-bearing: a sensor reported as selective for sarin has, in most cases, never been exposed to sarin.

What licenses the substitution is a claim of similarity. The compound chosen is said to be agent-like, and similarity is read from molecular descriptors: vapour pressure, dipole moment, frontier orbital energies, substructure and shape. The claim that matters for a detector is different. It is that the substitute produces the response the agent would produce in the same device. Those two claims are connected only if descriptor similarity transfers to response similarity, and that transfer has not been measured.

It has not been measured because measuring it requires the agents. Response data on authentic agents exist in few places, and the compounds used as simulants are rarely run alongside them. A recent survey of the field found no standard for chemical warfare agent simulants at all, and recorded that they have generally been selected on the basis of their

historical use.[7] Our institute is an OPCW designated laboratory, which identifies and assays authentic agents under an accredited procedure, so the comparison can be made on the agents themselves.

This work reports that measurement. A 29-dye colorimetric array was used to record the responses of seventeen authentic agents at five concentrations in two optical channels, and those responses were compared against the molecular-descriptor distances that simulant selection relies on. The array separates compounds in every run and channel, so the measurement is sound. Descriptor distance nonetheless carries no detectable information about response distance. Dimethyl methylphosphonate, the compound the field treats as the G-agent surrogate, is anti-correlated with two of the four G agents, and its resemblance to sarin weakens as the concentration falls towards the range where point detectors operate. Where the designation record and a regulatory toxicity model might have supplied the missing evidence, both are empty for the same compounds.

This is not a new sensing method, and we propose no replacement compound. It is a measurement of whether the substitutes the field already relies on carry the property they are relied upon for.

Results and Discussion

The array separates the agents

A comparison between descriptor space and response space is worth making only if the response measurement resolves the compounds at all. That question is already answered in the reports these data come from, and the answer does not depend on anything we do here. On the first run a linear discriminant classifier built on a six-dye subset under dual illumination separated the four G agents, VX and dimethyl methylphosphonate at 100% accuracy, against 87.5% under visible illumination alone.[8] On the second, a six-dye subset classified all seven agents at an accuracy of 0.964 with a macro F_1 of 0.964 under five-fold

stratified cross-validation.[9] On the third, a four-dye subset separated the four Novichok agents at an accuracy of 1.000 and a macro F_1 of 1.000 under leave-one-out cross-validation, over a working range of 10 to 500 μM with a detection limit near 10 μM.[10]

Two qualifications belong with those figures. Each is obtained on a dye subset chosen for that run, not on the 29-dye array we analyse, and the first run reports no held-out evaluation, so its 100% is an apparent rather than a validated accuracy. The two cross-validated figures are therefore the ones that carry the premise, and they carry it: an array that reaches 0.964 under five-fold and 1.000 under leave-one-out is resolving these compounds.

What our own check adds is narrower and is reported because the comparison that follows is made on patterns rather than on class labels. Each exposure gives a 29-dye ΔE_{00} vector, which we centred dye-wise at each concentration and normalised to unit length, so that agreement is measured between patterns and not between magnitudes, and took agreement as the cosine. The question it answers is whether a compound's pattern is stable enough across dose to be treated as that compound's pattern at all.

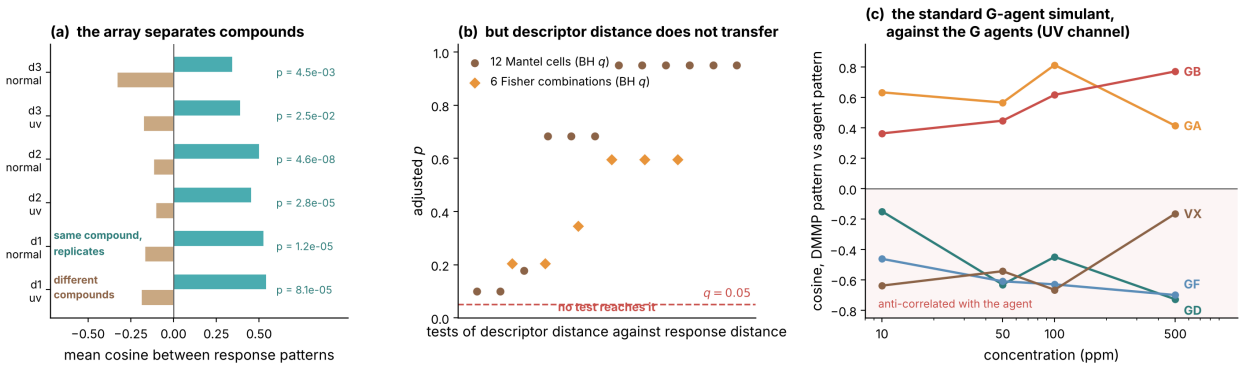


Figure 1: Response-space tests on authentic agents. (a) Mean cosine between response patterns, within one compound across concentrations and between different compounds, for three runs in two optical channels. (b) Adjusted p for every test of descriptor distance against response distance: twelve exact Mantel cells and six Fisher combinations of the independent runs. (c) Cosine between the response pattern of DMMP and those of the four G agents and VX, by concentration, in the ultraviolet channel.

It is stable enough (Figure 1a). One compound's patterns at two concentrations correlate at +0.345 to +0.542, different compounds anti-correlate at −0.103 to −0.330, and the

ordering holds in all six run-channel cells, with p from 4.6×10^{-8} to 0.025.

The tables analysed here hold one value per compound, concentration, dye and channel, so the within-compound term is agreement of a compound with itself at a different dose and not agreement between repeated measurements. Repeats were reported in the source work for the second and third runs, for selected dye-analyte combinations in the second[9] and for all experiments in the third,[10] but the repeat-level values are not in these tables, so no repeatability estimate enters any analysis below.

Two features of the measurement belong here rather than in the Supporting Information. The first is that dye-wise centring forces the mean between-compound cosine towards an algebraic floor of $-1/(n-1)$, and the observed means clear that floor by only 0.013 to 0.064 across the four cells where it could be evaluated. Most of the apparent separation is therefore a property of the centring, and what our test on its own establishes is the weaker statement that a compound agrees with itself across concentrations more than it agrees with other compounds; the classification results above, not this cosine, are what establish that the agents are separable. The second is that six dye columns in the ultraviolet pivot of the second run were bit-identical across every row. Tracing them to the source images showed that those dyes do not fluoresce under ultraviolet excitation and the regions of interest had read plate background, so one channel had been recorded six times. Columns identical across all rows are counted once, which leaves 23 effective channels in that cell and 25 in the ultraviolet pivot of the third run.

Descriptor distance does not predict response distance

Simulant selection rests on an implicit transfer: that a compound close to an agent in descriptor space will be close to it in whatever the detector measures. We tested the transfer directly, as the agreement between two distance matrices over the same compounds, one built from descriptors and one from response patterns. Agreement was measured by Mantel correlation and referred to a null built by enumerating every relabelling of the compounds,

so the test is exact rather than sampled.

The test is run within each run and never across runs. The three runs share no compound, so the dye-wise sensitivity of one plate cannot be calibrated against another, and pooling responses across them would require a commutability assumption we cannot support. That restriction sets the power. Six compounds in the first run and six in the second carry both a response pattern and a descriptor vector, giving fifteen pairs in each, and the third run drops out because the A-series compounds are not in the descriptor panel. Nitrogen mustard drops out of the second run for the same reason.

Across the twelve cells, three spaces by two channels by two runs, nothing survives correction (Figure 1b). Three cells reach a nominal p below 0.05, all of them in the second run: the merged space in the ultraviolet channel at 0.0153, the structural space in the same channel at 0.0167, and the electronic space in the reflectance channel at 0.0444. Benjamini–Hochberg correction over the twelve leaves a smallest adjusted value of 0.10. The pattern of the three is itself uninformative about transfer, because the two ultraviolet cells implicate the structural and merged spaces while the reflectance cell implicates the electronic space, and the same space does not reproduce across channels of the same plate.

Combining runs does not change the answer. The first and second runs share neither compounds nor plates, so their exact Mantel p values are independent and can be combined by Fisher's method. We report all six combinations of three spaces and two channels rather than selecting one, which would reintroduce the analyst freedom the test is meant to remove. The smallest adjusted value over the six is 0.204, and none reaches 0.05.

What this does not license is the claim that no transfer exists. Fifteen pairs is a weak test, and the pooled rank correlations are correspondingly imprecise: the structural space in the ultraviolet channel gives $\rho = +0.31$ with a 95% interval from -0.45 to $+0.81$, which excludes neither zero nor a strong association. The defensible statement is that the transfer on which simulant selection depends has not been established, in the one body of authentic-agent response data where it could be tested, and that the experiment which would settle it

is a run containing a compound already measured in an existing run. Such a bridge would fix the plate correction and open the 136 cross-run pairs that are closed today.

The standard G-agent simulant against the G agents

The transfer test above treats all compounds alike. One compound deserves to be named, because the field treats it as the G-agent surrogate and because decontamination efficiency is quantified against it. The first run contains DMMP together with the four G agents and VX, so its response can be set against theirs directly.

In the ultraviolet channel DMMP correlates with tabun at $+0.764$ and with sarin at $+0.645$, and anti-correlates with soman at -0.492, cyclosarin at -0.721 and VX at -0.702. In the reflectance channel the same comparison gives -0.112, $+0.569$, -0.479, -0.495 and -0.273. The mean over the four G agents is negative in both channels, -0.189 and -0.135. The compound used in place of the G agents does not, on this measurement, resemble the G agents as a class.

A weaker claim survives in one channel. If DMMP is G-like at all it should sit closer to the G series than to VX, and that contrast is $+0.513$ with a 95% interval from $+0.131$ to $+0.722$ in the ultraviolet channel, which excludes zero. In the reflectance channel the same contrast is $+0.137$ with an interval from -0.144 to $+0.290$, which does not. One optical channel supports the ordering and the other does not resolve it.

Concentration changes the picture in the direction that matters for a detector (Figure 1c). The DMMP-to-sarin cosine rises monotonically from $+0.363$ at 10 μM to $+0.771$ at 500 μM, and the DMMP-to-cyclosarin cosine falls monotonically from -0.462 to -0.700 over the same range. Monotonicity is perfect in three of the ten agent-channel series. It is also the strongest result the design can give: with four concentrations the two-sided exact permutation has a floor of $p = 0.083$, three series reach exactly that floor, and Benjamini–Hochberg correction over the ten leaves a smallest adjusted value of 0.278. We therefore report the direction and not a significance claim. The direction is that the resemblance to sarin is weakest at the low

end of the range, which is the end at which point detectors are expected to alarm.

Two further readings suggested themselves to us and did not survive their own controls. We report both, because the alternative is to present artefacts as results.

The first was a ranking. DMMP is the nearest of the five candidates to tabun in the ultraviolet channel and to sarin in the reflectance channel, and it is nearest to nothing else, and the two channels never agree on which agent it is nearest to. That reads as evidence that the choice of optical channel decides which agent the standard simulant stands in for. It is not. With five candidates the probability of rank one is 1/5 under random assignment, the expected number of first places over ten agent-channel cells is 2.0, and the observed number is two, with a one-sided binomial p of 0.62. The ranking carries no information.

The second was a channel decomposition. The cosine is a sum over dye channels, and the DMMP-to-tabun value of +0.764 turns out to rest on two of the 29 channels, with the largest single channel carrying 44% of the positive mass. Read alone, that makes the resemblance look fragile. Two controls show it is not a property of DMMP. Agent-to-agent pairs in the same run concentrate identically, with median two channels in the ultraviolet and one in the reflectance channel, and a null built by permuting the dye labels of the DMMP vector gives a mean of 2.00 and 2.26. Concentration of this kind is what the product of two centred unit vectors in 29 dimensions does, and it says nothing about this compound.

What remains after both controls is the set of measured values themselves, and one structural fact. The first run contains no candidate simulant other than DMMP, so the designation cannot be contrasted against an alternative in response space at all. The permutation null that the descriptor analysis of the next section relies on is, in response space, empty.

Descriptor space cannot supply the missing validation

If response space cannot establish the transfer, the question becomes whether descriptor space can at least establish the premise it is used for, that the incumbents sit closer to the

agents than other compounds do. This can be asked exactly. Designating three compounds out of the non-agents in a panel is a choice among a finite set, so the incumbent designation can be referred to every alternative designation rather than to a sampled null. The primary panel of thirty-five compounds gives 2024 designations and the reduced panel of twenty-eight gives 680.

On the primary panel the answer is flat (Figure 2). No agent of the eleven has a valid match, the smallest nominal p over the eighteen tests is 0.1695, and the Benjamini–Hochberg adjusted value is 0.6173. Nothing in that panel supports the designation.

The reduced panel, which omits five compounds the literature also calls simulants, reads differently. Four agents acquire a valid match, and the merged space returns a nominal one-sided p of 0.0250 with the structural space at 0.0265. Whether that survives depends on which correction is applied. Benjamini–Hochberg over the eighteen tests gives 0.1809, correcting within each space gives 0.0824, and the Westfall–Young single-step procedure, which uses the shared permutations and so handles the dependence between tests exactly, gives 0.0779. Only the scheme that fixes one statistic in advance returns a value below 0.05, at 0.0397. Significance here is a property of one correction scheme on one panel construction, and we report all four rather than the one that passes.

The surviving structural value is also an ordinary draw from the choices available to an analyst. Drawing three descriptors at random from the structural block 2000 times and repeating the whole procedure places the observed value at the 24.3rd percentile of that distribution. A value at the quarter mark of what random descriptor choices produce is not evidence about chemistry.

One quantity in this family does separate the incumbents from the alternatives, and it is worth stating precisely because it is easy to over-read. Across 500 random three-descriptor subspaces, the incumbent designation reaches $p < 0.05$ in 36.0% of them. Referred to the same quantity computed for all 680 designations, where the mean is 4.7% and the median 2.0%, the incumbent designation ranks fifth of 680 ($p = 0.0074$). The incumbents are

therefore unusual in how often a lucky choice of three descriptors can be found for them. That is a statement about the flexibility of the comparison, not about proximity to the agents.

What decides the answer is panel composition. Adding back the five compounds the sensing literature also treats as simulants moves the count from four valid matches to zero, because diisopropyl fluorophosphate is nearer to three G agents than any incumbent is, and 2-chloroethyl ethyl sulfide is nearer to sulfur mustard than dichloromethane is. The incumbents do not move. What moves is whether anything closer is allowed into the comparison.

Neither the designation record nor a toxicity model fills the gap

Two places remain where the missing evidence might be found, and both are empty for the same compounds.

The first is the standard-setting record. We traced each incumbent to the document that designates it. DMMP is identified in a NIOSH programme for respirator materials resistance testing, where the measurement is filter breakthrough and the same document states that the simulant information is not to be used as a recommendation of any simulant.[3] DPGME appears in a US Army patent for a training composition, where the property required of it is detectability by ion mobility spectrometry.[4] For dichloromethane we found no designating document in the public record. Each of the two traceable designations is a validation against a measurement, and neither of those measurements is detection. A simulant is validated for a purpose, and that validation does not travel with the compound to a different purpose.

The second is regulatory toxicity (Figure 3). One of the three incumbents carries a regulator-derived acute inhalation value. DMMP, the compound against which decontamination efficiency is quantified, carries none of any kind. A model cannot fill that gap. A random forest trained on 230 AEGL chemicals reaches $R^2 = 0.458$ in cross-validation, with an out-of-fold bias of $+0.001$ log units, so nothing in its internal statistics warns a user off. Applied to twenty military agents held out of training it is wrong in one direction without

exception, by 0.40 to 3.42 log units. Its applicability-domain check certifies fourteen of the twenty as inside the domain, and the worst of those fourteen is wrong by 2.68 log units, a factor of 482. The mean error over the twenty is +1.68 log units with Cohen's $d_z = +1.96$.

Conclusions

Simulant validity is not a property of a compound. It is a claim about a measurement, and the practice of designating chemical warfare agent simulants records neither the measurement a compound was validated against nor the comparison in which its validity was tested.

Three counts fix where the evidence stands. Across the seventeen authentic agents measured here, exactly one designated simulant has ever been measured beside them, and that single co-measurement sits in one run of one array. Twelve exact Mantel tests and six combinations of them leave no agreement between descriptor distance and response distance below $q = 0.05$. On the primary panel no agent of the eleven has a valid match to an incumbent simulant, at a Benjamini–Hochberg adjusted value of 0.6173.

The documentary record accounts for the first of those counts rather than contradicting it. DMMP was designated for respirator filter breakthrough testing and DPGME for an ion-mobility training composition. Neither measurement is detection, and a validation does not travel with a compound to a purpose it was never tested for. The toxicity axis is empty in the same place: DMMP, the compound against which decontamination efficiency is quantified, carries no regulator-derived acute inhalation value, and a random forest trained on 230 AEGL chemicals is wrong in one direction without exception on twenty held-out military agents, so a prediction cannot stand in for the missing measurement.

Absence of evidence is not evidence of unsuitability, and we conclude nothing about whether the incumbents are poor choices. We propose no replacement compound either, because selecting one on descriptor similarity would repeat the error this paper is about.

What can be specified is the measurement that would settle the question. It is a bridging

run: candidate simulants and the agents they stand in for exposed on the same plate, in the same optical channels, at the same concentrations, within one session. Such a run yields the one quantity no amount of descriptor work can supply, a response-space distance between a simulant and its agent measured under identical conditions, and it needs no new chemistry and no new instrument. It needs only that the simulant be placed beside the agent rather than in place of it. Until that run exists, a statement that a compound is a validated detection simulant reports a convention, and a convention should be declared as one.

Experimental Section

Array measurements

The response data are the colorimetric tables of three published runs of one array, reanalysed here and not remeasured. The first covers tabun, sarin, soman, cyclosarin, VX and dimethyl methylphosphonate;[8] the second hydrogen cyanide, cyanogen chloride, phosgene, chloropicrin, sulfur mustard, nitrogen mustard and lewisite;[9] the third the Novichok agents A-230, A-232, A-234 and A-242.[10] All agents were produced and handled at the CBRN Defense Research Institute, Seoul, an Organisation for the Prohibition of Chemical Weapons designated laboratory, under Chemical Weapons Convention procedures. Purity was confirmed by GC–MS and exceeded 95% for every compound of the second run except lewisite, at 67%.

The array is the same 29 commercial chromogenic and fluorogenic dyes in the same order in all three runs, commercially sourced and used as received. Stock solutions at 100 mM were prepared in ethanol, methanol, dimethyl sulfoxide, water, tetrahydrofuran or toluene and sonicated to dissolution, then diluted into acetonitrile to a working concentration of 100 μM, which holds the residual stock solvent below 0.1% (v/v) and places every dye in one solvent environment. Each analyte was delivered in acetonitrile by serial dilution to final concentrations of 0, 10, 50, 100 and 500 μM in a 2.0 mL dye solution. Samples were mixed without incubation and imaged at once, under visible light and under ultraviolet excitation

at 365 nm, with a smartphone camera fixed in an enclosed imaging box at 22 ± 3 °C and 35 ± 5% relative humidity. Regions of interest were central 20 × 10 pixel zones; RGB values were read at five positions within each region, averaged, converted to CIE $L^*a^*b^*$, and expressed as the CIEDE2000 colour difference ΔE_{00} against the 0 μM image of the same dye in the same channel. The self-referencing removes illumination and device offsets common to an image pair.

The three runs are different compounds measured on different plates and are not replicates of one another, so no ΔE_{00} value is compared across runs; every analysis below is run within a run and channel, and results are combined only at the level of correlation coefficients. The two channels are physically different measurements rather than repeats and are treated as independent modalities. The tables hold one value per compound, concentration, dye and channel.

Response patterns

Each exposure gives a 29-dye ΔE_{00} vector. Vectors were centred dye-wise within each concentration and normalised to unit length, so that agreement is measured between patterns and not between magnitudes, and agreement was taken as the cosine. Dye columns that are bit-identical across every row carry no information and were counted once, which leaves 23 effective channels in the ultraviolet pivot of the second run and 25 in that of the third.

Descriptors

Structural and physicochemical descriptors, 51 after excluding run metadata, were computed with RDKit from canonical SMILES. Ten are read from a three-dimensional structure and are taken as Boltzmann-weighted means over an ETKDGv3 conformer ensemble at 298 K with MMFF94 energies, or UFF for lewisite 1, the one compound without MMFF parameters, so that no result depends on an undeclared choice of conformer. Electronic descriptors were computed with GFN2-xTB in the gas phase, Boltzmann averaged over up to five conformers

at 298.15 K. The electronic space carries E_{HOMO}, E_{LUMO} and the dipole moment. The Parr electrophilicity index $\omega = \mu^2/2\eta$ is reported but admitted to no distance, because it is an exact function of the two orbital energies that accompany it and a filter built on linear correlation does not detect that. Dipole moments were recomputed at B3LYP/def2-SVPD to confirm that the reported result does not depend on the semiempirical method. Identifiers and formulae were resolved against PubChem, and no property value was taken from a prediction service.

Distances and tests

Within each space, descriptors of zero variance were removed, then one of each pair correlated at $|r| > 0.95$, retaining a protected set. Remaining descriptors were standardised and distances taken as Euclidean, normalised by the median nearest-neighbour separation of the same space, which makes a normalised distance comparable within a panel and not between panels.

A paired comparison of the distance to the nearest incumbent simulant against the distance to the nearest interferent is confounded by group size, since the minimum of many draws is smaller than the minimum of few by construction. We therefore enumerated every way of designating three of the non-agents as simulants and report the rank of the actual designation as an exact p-value: $\binom{24}{3} = 2024$ designations on the primary panel and $\binom{17}{3} = 680$ on the twenty-eight-compound panel. The hypothesis is directional, so the reported p is one-sided; the two-sided conventions are given alongside it. A valid match requires a normalised distance of 2 or less together with a distance smaller than that to every compound outside the designated set. Sensitivity to descriptor choice was assessed by resampling the merged-space descriptors with replacement 1000 times under a fixed seed, restandardising and recomputing within each replicate; the intervals so obtained are percentile ranges over descriptor choice and not confidence intervals in the sampling sense.

Agreement between descriptor distance and response distance was tested by Mantel corre-

lation with exact enumeration of relabellings in each run-channel cell, with the independent runs combined by Fisher's method and the family of tests corrected by the Benjamini–Hochberg procedure. Six distance statistics and six multiplicity schemes were carried through in parallel so that the conclusion can be read against the analyst choices rather than from one of them.

Software

NumPy 2.5.3, SciPy 1.18.1, pandas 3.0.5, scikit-learn 1.9.1, Matplotlib 3.11.2, RDKit 2026.03.6 and PySCF 2.14.0. All analyses are deterministic given the seed, and the permutation tests are exhaustive rather than sampled. Code, descriptor tables and figure scripts are in the repository cited below.

Acknowledgement

Supporting Information Available

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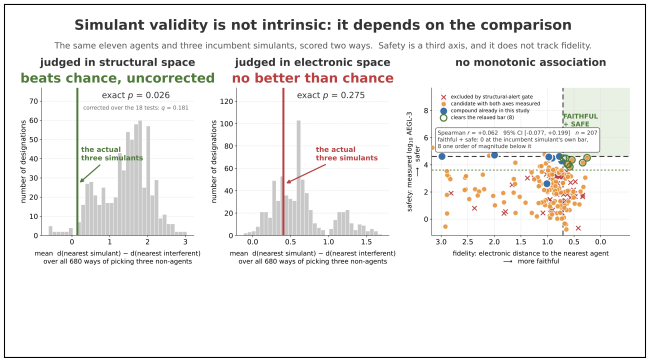
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TOC Graphic


Simulant validity is not intrinsic: it depends on the comparison
The same eleven agents and three incumbent simulants, scored two ways. Safety is a third axis, and it does not track fidelity.
judged in structural space
beats chance, uncorrected
exact p = 0.026
corrected over the 18 tests: q = 0.181
the actual three simulants
number of designations
mean d(nearest simulant) − d(nearest interferent) over all 680 ways of picking three non-agents
judged in electronic space
no better than chance
exact p = 0.275
the actual three simulants
number of designations
mean d(nearest simulant) − d(nearest interferent) over all 680 ways of picking three non-agents
no monotonic association
excluded by structural-alert gate
candidate with both axes measured
compound already in this study
clears the relaxed bar (8)
FAITHFUL + SAFE
Spearman r = +0.062 95% CI [-0.077, +0.198] n = 207
faithful + safe: 0 at the incumbent simulants own bar,
8 one order of magnitude below it
safety: measured log10 AEGL-3
fidelity: electronic distance to the nearest agent
→ more faithful

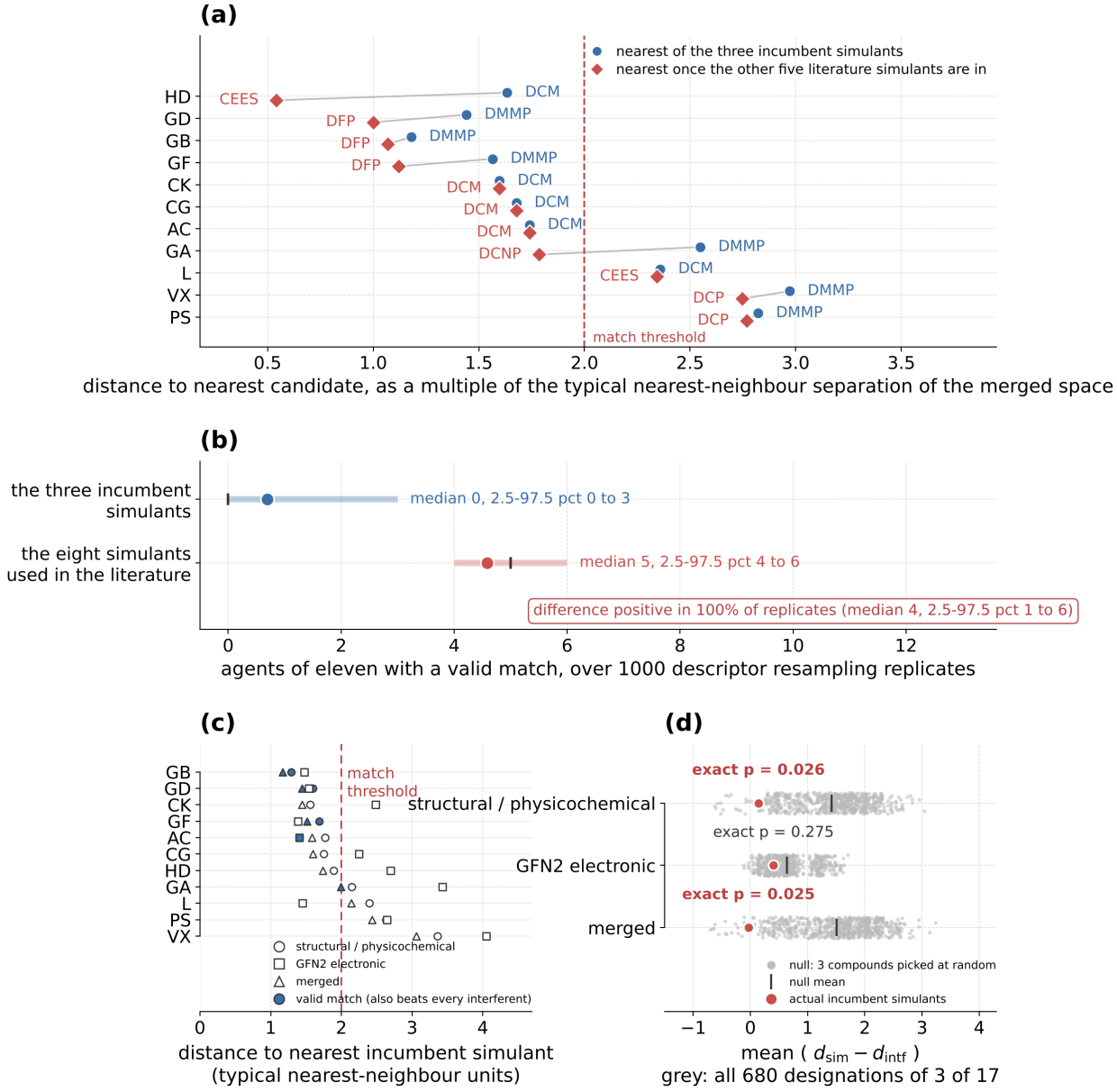


Figure 2: Descriptor-space tests. (a) Distance from each agent to its nearest candidate, for the three incumbents and for the eight compounds the sensing literature also calls simulants. (b) Valid matches over 1000 descriptor resampling replicates. (c) Distance to the nearest incumbent by space. (d) Observed mean difference against the exact null over all 680 designations of three compounds from seventeen.

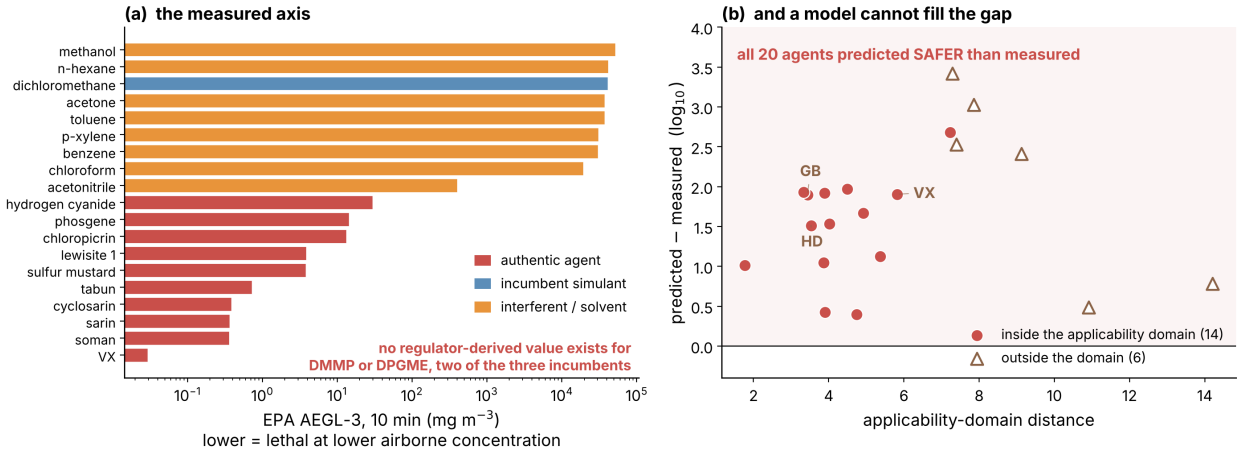


Figure 3: The toxicity axis. (a) Measured EPA AEGL-3 10-minute values for the panel compounds that carry one. (b) Prediction error of a random forest trained on 230 AEGL chemicals, for twenty military agents held out of training, against distance to the training set.